BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Boolean Conditions in Code

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What are 17 DIV 5 and 17 MOD 5?

2. What happens in the fetch stage?
   A) Work out its meaning
   B) Get the next instruction
   C) Carry out the instruction
   D) Save a backup copy

3. "Execution only copies an instruction into memory without performing it." Correct this. Explain why.

4. A CPU compares scores and adds a bonus. Correct both claims. Explain why. (a) The Control Unit stores documents forever. (b) The ALU stores the next instruction's address.

Today's objectives

Use comparisons and Boolean operators in conditions. Use boolean and and in a practical scenario.

Key vocabulary - Read, cover, recall

Boolean
A Boolean value represents true or false.

AND
An AND condition is true only when both component conditions are true.

OR
An OR condition is true if at least one condition is true.

NOT
NOT reverses a Boolean value, changing true to false or false to true.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

What is Boolean?

Boolean - A Boolean value represents true or false.

In this example

A swimming-pool entry policy uses and; An AND condition is true only when both component conditions are true.

Follow the effect

An OR condition is true if at least one condition is true. Explain how this changes the eventual program result.

Check understanding

NOT is used for a specific reason: NOT reverses a Boolean value, changing true to false or false to true.

I Do - Worked model

A swimming-pool entry policy is being written. Explain how boolean and and are used to make the program work. [4 marks]

- We are designing a swimming-pool entry policy. Identify which data and decisions are essential.
- Boolean [1] - A Boolean value represents true or false. [1].
- AND [1] - An AND condition is true only when both component conditions are true. [1].
- Two linked points for each technique give four mark points.

We Do - Guided practice

1. In a club membership check, explain how boolean helps the program. [2 marks]

2. For a alarm controller, explain why and matters. [2 marks]

3. When designing a timetable access rule, explain how or and not should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean expression so the result is false when only one check passes.

```
hasPassword = true
hasSecondFactor = false
print(hasPassword ... hasSecondFactor)
```

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A gym opening condition needs to work correctly. Explain how and and or would help. [4 marks]

Q4. A pupil is programming a museum visitor rule. Explain the roles of boolean, or and not in their solution. [6 marks]

Exit Ticket - True or False?

A Boolean value represents true or false.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. DIV gives integer quotient 3; MOD gives remainder 2.

2. B - Get the next instruction

3. The operation specified by the decoded instruction is carried out.

4. (a) The Control Unit decodes instructions and controls CPU operations. (b) The ALU handles arithmetic and logic. Registers store data or addresses briefly.

I Do - Explain the model

We are designing a swimming-pool entry policy. Identify which data and decisions are essential.

Boolean [1] - A Boolean value represents true or false. [1].

AND [1] - An AND condition is true only when both component conditions are true. [1].

Two linked points for each technique give four mark points.

We Do - Model answers

1. Boolean [1] - A Boolean value represents true or false. [1].

2. AND [1] - An AND condition is true only when both component conditions are true. [1].

3. OR [1] - An OR condition is true if at least one condition is true. [1]. NOT [1] - NOT reverses a Boolean value, changing true to false or false to true. [1].

Code First - Answer

```
hasPassword = true
hasSecondFactor = false
print(hasPassword AND hasSecondFactor)
Expected output: false
```

You Do - Self-mark

Q2: Boolean [1]; OR [1].

Q3: AND [1] - An AND condition is true only when both component conditions are true. [1]. OR [1] - An OR condition is true if at least one condition is true. [1].

Q4: Boolean [1] - A Boolean value represents true or false. [1]. OR [1] - An OR condition is true if at least one condition is true. [1]. NOT [1] - NOT reverses a Boolean value, changing true to false or false to true. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.